

Projects

Continuous Integration & Deployment Pipeline (2025)

Tech Stack: Jenkins, GitHub, Docker

- Designed and implemented a **7-stage CI/CD pipeline** for automated build, test, and deployment workflows. - Integrated GitHub version control and automated triggers for faster releases. - Implemented error handling and email notifications, improving deployment reliability. - Strengthened understanding of **production-ready SaaS deployment pipelines**.

Parallel Matrix Multiplication (2025)

Tech Stack: C++, pthreads, OpenMP

- Built sequential and parallel implementations to analyze performance bottlenecks. - Optimized computation using multithreading, achieving significant speedup on multi-core systems. - Gained hands-on experience with **scalability and algorithm optimization**.

DNS Server Implementation (2025)

Tech Stack: Python, UDP Sockets

- Implemented a DNS server handling **A and CNAME records** using UDP-based communication. - Added logging, validation, and error handling to improve reliability. - Strengthened fundamentals of **networking and distributed coordination**.

Staff Data Management System (2024)

Tech Stack: HTML, CSS, JavaScript, Firebase

- Developed a responsive web application for staff registration and verification. - Implemented **real-time database synchronization and authentication** using Firebase. - Improved data consistency and user experience across sessions.

Theft-Free Cart – Smart Retail System (2023)

Tech Stack: Arduino Nano 33 IoT, RFID, IR Sensor, Load Cell, MQTT, Firebase

- Built a real-time inventory and billing system integrating IoT devices with cloud services. - Reduced checkout delays by **30%** through automation and sensor optimization. - Demonstrated end-to-end ownership from hardware integration to backend data processing.

Certifications

- **Design Create Test (DCT) – 2024**
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Soft Skills

- **Ownership & Accountability:** Delivered complete projects from design to deployment.
 - **Problem Solving:** Debugged complex systems and optimized performance across projects.
 - **Collaboration:** Worked closely with team members using Agile/Scrum practices.
 - **Communication:** Presented technical projects clearly to academic and technical audiences.
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Languages

- **English:** Fluent (C2 – Written & Spoken)